

Project: AdStudio

Digital Advertising & Campaign Management Platform

Domain: Communications, Media & Technology

1. Introduction

AdStudio is a web-based digital advertising and campaign management platform designed for advertising agencies, media buying firms, brand marketing teams, and publishers to manage advertiser accounts, campaign planning, ad creative management, media plan execution, performance tracking, billing and reconciliation, and campaign analytics. The system enables advertisers to manage their brand portfolio and campaign briefs, media planners to create and execute media plans, creative teams to manage ad assets, finance teams to track billing and vendor payments, and publishers to manage their ad inventory and insertion orders.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **Advertiser / Brand Manager:** Submits campaign briefs, reviews media plans, and tracks campaign performance
- **Media Planner:** Creates media plans, allocates budgets across channels, and places insertion orders
- **Creative Manager:** Uploads and manages ad creatives, tracks approval status, and links assets to campaigns
- **Publisher / Media Owner:** Manages ad inventory, responds to insertion orders, and submits delivery reports
- **Finance Executive:** Manages client billing, vendor invoices, reconciliation, and payment tracking
- **Ad Operations Admin:** Configures channel catalog, publisher inventory, rate cards, and user roles

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.
- **2.2 Advertiser & Brand Management:** Manages advertiser profiles, brand portfolios, and budget allocations.
- **2.3 Campaign Planning & Briefing:** Manages campaign briefs, objectives, target audience definitions, and approval workflows.
- **2.4 Media Plan & Insertion Order Management:** Creates media plans across channels, generates insertion orders, and tracks publisher confirmations.
- **2.5 Creative Asset Management:** Manages ad creative uploads, specifications validation, approval workflows, and version control.
- **2.6 Campaign Delivery & Performance Tracking:** Records delivery data, tracks KPIs against planned targets, and manages performance exceptions.
- **2.7 Billing, Reconciliation & Vendor Management:** Manages client billing, publisher invoice reconciliation, and payment processing.
- **2.8 Campaign Analytics & Reporting:** Provides dashboards for impressions, reach, engagement, spend pacing, and ROI metrics.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- Advertiser, agency, and publisher staff registration with role-based access control
- Account-scoped data visibility and full audit logging for all campaign financial actions

Entities

- **User**

- UserID
- Name
- Role (AdvertiserBrand/MediaPlanner/CreativeManager/Publisher/Finance/Admin)
- Email
- Phone
- AccountID
- Status (Active/Inactive/Suspended)
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - EntityType
 - Timestamp

4.2 Advertiser & Brand Management Module

Features

- Register advertiser accounts with brand portfolios and annual budget allocations
- Track budget consumption and remaining headroom across active campaigns

Entities

- **Advertiser**
 - AdvertiserID
 - CompanyName
 - Industry
 - AccountManagerID
 - AnnualBudget
 - Currency

- Status (Active/Inactive/Suspended)
- **Brand**
 - BrandID
 - AdvertiserID
 - BrandName
 - Category
 - AllocatedBudget
 - SpentToDate
 - Status (Active/Discontinued)

4.3 Campaign Planning & Briefing Module

Features

- Accept campaign briefs with objectives, target demographics, and channel preferences
- Manage internal approval workflows before activating media planning

Entities

- **CampaignBrief**
 - BriefID
 - BrandID
 - CampaignName
 - Objective (Awareness/Consideration/Conversion/Retention)
 - TargetDemographic
 - Geography
 - StartDate
 - EndDate
 - TotalBudget
 - ChannelPreferences

- SubmittedByID
- Status (Draft/Submitted/Approved/Rejected/Active/Completed)
- **TargetAudience**
 - AudienceID
 - BriefID
 - AgeRange
 - Gender
 - Interests
 - Geography
 - DeviceType (Desktop/Mobile/Tablet/CTV/All)
 - Status (Active/Archived)

4.4 Media Plan & Insertion Order Management Module

Features

- Build multi-channel media plans with budget allocation, flight dates, and KPI targets
- Generate insertion orders, send to publishers, and track confirmation status

Entities

- **MediaPlan**
 - PlanID
 - BriefID
 - PlannerID
 - TotalBudgetAllocated
 - ChannelMix
 - StartDate
 - EndDate
 - Status (Draft/PendingApproval/Approved/Active/Completed)

- **MediaLineItem**
 - LineItemID
 - PlanID
 - Channel (Display/Video/Social/Search/OOH/Print/Radio)
 - Publisher
 - Format
 - PlannedImpressions
 - PlannedBudget
 - CPM
 - FlightStart
 - FlightEnd
 - Status (Planned/Ordered/Live/Paused/Completed)
- **InsertionOrder**
 - IOID
 - LineItemID
 - PublisherID
 - OrderDate
 - StartDate
 - EndDate
 - CommittedImpressions
 - OrderValue
 - Status (Sent/Confirmed/Rejected/Delivered/Disputed)

4.5 Creative Asset Management Module

Features

- Upload and manage ad creatives with format, dimension, and file size specifications

- Track creative approval workflows and link approved assets to media line items

Entities

- **CreativeAsset**
 - AssetID
 - BrandID
 - CampaignBriefID
 - AssetName
 - Format (Banner/Video/Native/Text/Audio/Rich Media)
 - Dimensions
 - FileSizeKB
 - FilePath
 - Version
 - UploadedByID
 - Status (Draft/PendingApproval/Approved/Rejected/Archived)
- **CreativeApproval**
 - ApprovalID
 - AssetID
 - ReviewerID
 - ReviewDate
 - Decision (Approved/Rejected/RevisionRequired)
 - Feedback
 - Status (Pending/Completed)
- **AssetLineItemLink**
 - LinkID
 - AssetID
 - LineItemID

- LinkedDate
- Status (Active/Replaced/Removed)

4.6 Campaign Delivery & Performance Tracking Module

Features

- Record delivery data from publishers against planned impressions and budget
- Track pacing, identify under-delivery, and manage make-good arrangements

Entities

- **DeliveryRecord**
 - DeliveryID
 - LineItemID
 - IOID
 - ReportingDate
 - DeliveredImpressions
 - Clicks
 - Spend
 - Source (PublisherReport/InternalEntry)
 - Status (Accepted/Disputed/PendingVerification)
- **PacingAlert**
 - AlertID
 - LineItemID
 - AlertType (UnderDelivery/OverDelivery/BudgetExhausted/FlightEndApproaching)
 - AlertDate
 - PacingPercent
 - Status (Open/Actioned/Closed)

4.7 Billing, Reconciliation & Vendor Management Module

Features

- Generate client invoices from approved delivery data and manage payment collection
- Reconcile publisher delivery reports against insertion orders and process vendor payments

Entities

- **ClientInvoice**
 - InvoiceID
 - AdvertiserID
 - CampaignBriefID
 - BillingPeriod
 - InvoiceAmount
 - AgencyCommission
 - NetBillable
 - IssuedDate
 - Status (Draft/Issued/Paid/Disputed/Overdue)
- **PublisherInvoice**
 - PubInvoiceID
 - PublisherID
 - IOID
 - InvoiceAmount
 - DeliveredValue
 - VarianceAmount
 - ReceivedDate
 - Status (Received/Reconciled/Discrepancy/Paid)

4.8 Campaign Analytics & Reporting Module

Features

- Impression, spend pacing, CPM, and CTR dashboards by campaign, channel, and brand
- Publisher delivery performance, creative effectiveness comparison, and billing reconciliation summary

Entities

- **CampaignReport**
 - ReportID
 - Scope (Brand/Campaign/Channel/Publisher/Period)
 - Metrics (TotalImpressions, TotalClicks, CTR, TotalSpend, CPM, CPC, DeliveryRate, BudgetPacingPercent, AgencyROI)
 - GeneratedDate

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB
- **Production:** Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, Name, Role, Email, Phone, AccountID, Status)
- **AuditLog**(AuditID, UserID, Action, EntityType, Timestamp)
- **Advertiser**(AdvertiserID, CompanyName, Industry, AccountManagerID, AnnualBudget, Currency, Status)
- **Brand**(BrandID, AdvertiserID, BrandName, Category, AllocatedBudget, SpentToDate, Status)
- **CampaignBrief**(BriefID, BrandID, CampaignName, Objective, TargetDemographic, Geography, StartDate, EndDate, TotalBudget, ChannelPreferences, SubmittedByID, Status)

- **TargetAudience**(AudienceID, BriefID, AgeRange, Gender, Interests, Geography, DeviceType, Status)
- **MediaPlan**(PlanID, BriefID, PlannerID, TotalBudgetAllocated, ChannelMix, StartDate, EndDate, Status)
- **MediaLineItem**(LineItemID, PlanID, Channel, Publisher, Format, PlannedImpressions, PlannedBudget, CPM, FlightStart, FlightEnd, Status)
- **InsertionOrder**(IOID, LineItemID, PublisherID, OrderDate, StartDate, EndDate, CommittedImpressions, OrderValue, Status)
- **CreativeAsset**(AssetID, BrandID, CampaignBriefID, AssetName, Format, Dimensions, FileSizeKB, FilePath, Version, UploadedByID, Status)
- **CreativeApproval**(ApprovalID, AssetID, ReviewerID, ReviewDate, Decision, Feedback, Status)
- **AssetLineItemLink**(LinkID, AssetID, LineItemID, LinkedDate, Status)
- **DeliveryRecord**(DeliveryID, LineItemID, IOID, ReportingDate, DeliveredImpressions, Clicks, Spend, Source, Status)
- **PacingAlert**(AlertID, LineItemID, AlertType, AlertDate, PacingPercent, Status)
- **ClientInvoice**(InvoiceID, AdvertiserID, CampaignBriefID, BillingPeriod, InvoiceAmount, AgencyCommission, NetBillable, IssuedDate, Status)
- **PublisherInvoice**(PubInvoiceID, PublisherID, IOID, InvoiceAmount, DeliveredValue, VarianceAmount, ReceivedDate, Status)
- **CampaignReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Advertiser Portal:** Brand dashboard, campaign brief submission form, media plan review, delivery performance view
- **Media Planner Console:** Media plan builder, line item scheduler, insertion order generator, pacing alert board
- **Creative Studio:** Asset upload form, approval workflow tracker, asset-to-campaign linker, version history
- **Publisher Portal:** Insertion order inbox, confirmation form, delivery report submission, invoice submission

- **Finance Dashboard:** Client invoice generator, publisher invoice reconciliation, payment tracker, billing calendar
- **Admin Console:** Channel catalog, publisher registry, rate card management, campaign product and user configuration

8. Non-Functional Requirements

- **Performance:** Support 8,000+ concurrent users across advertisers, planners, and publishers during campaign launch peaks
- **Security:** Advertiser budget data access control, creative asset rights management, RBAC, and full audit trails
- **Scalability:** Multi-advertiser and multi-publisher architecture supporting large agency holding group deployments
- **Availability:** 99.9% uptime with campaign delivery tracking and billing modules prioritised for high availability
- **Maintainability:** Configurable channel types, creative format specifications, rate cards, and billing cycle rules
- **Observability:** Real-time campaign pacing dashboards, delivery discrepancy alerts, and billing reconciliation monitoring

9. Assumptions & Constraints

- Phase 1 excludes integration with programmatic DSP/SSP platforms, ad servers, and third-party measurement tools
- Delivery data is entered from publisher reports manually; automated ad server feed integration is deferred to Phase 2
- All notifications are in-app only; email delivery reports and Slack workflow integrations are out of Phase 1 scope
- Creative asset files are stored as local file path references; cloud CDN asset hosting integration is deferred
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling